Validity Is Not Difficulty: Hazard-Preserving Physical 3D World Compilation for Autonomous Driving

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Abstract

*Driving-world reconstruction systems often conflate two fundamentally different causes of difficulty: a scene may be hard because it contains a legitimate hazardous actor, or invalid because its reconstructed geometry contradicts sensor evidence. Treating both cases as low confidence can obtain superficially safer worlds by deleting precisely the actors that matter. We introduce HARP-3D, a hazard-preserving physical 3D world compiler that separates appearance from collision geometry and artifact validity from behavioral hazard. The compiler aligns multi-frame Actor evidence in canonical coordinates and exposes four explicit surface actions—*KEEP, PROJECT, COMPLETE, *and* UNKNOWN*—without changing Actor identity, trajectory, or extent. A paired factorization objective holds artifact predictions invariant to legal hazard interventions and hazard predictions invariant to geometry artifacts. Finally, learned Actor residual distributions are combined with analytic trajectory-boundary geometry to produce a continuous, multi-horizon task-cost density and a runtime reliability surface that is monotone in both cost budget and horizon. On a frozen zero-shot Argoverse 2 cohort, ray-certified compilation eliminates paired free-space violations, reduces target LiDAR depth error from 0.207 to 0.154 m and symmetric Chamfer from 0.254 to 0.175 m, and retains all Actor and hazard state. A nearest-surface alternative attains 0.168 m Chamfer but leaves 84.9% matched-ray violations, exposing an explicit geometry–safety boundary. Remaining factorization and reliability values stay frozen until their milestones complete.*

1. Introduction

Reconstructed driving worlds are increasingly used for perception analysis, counterfactual replay, and planning evaluation. Their geometry, however, is rarely a literal collision model. A rendered Gaussian may explain image appearance while occupying free space, duplicating an Actor shell, or flickering across time. Conversely, a perfectly valid Actor may be difficult because it cuts in, brakes hard, or crosses the Ego path. A useful world compiler must distinguish these cases: *world validity is not world difficulty*.

This distinction matters because a generic confidence filter has an easy but scientifically undesirable failure mode. It can improve aggregate collision proxies by suppressing uncertain or nearby Actors. Such a world looks safer because legitimate hazards disappeared, not because its physical state became more faithful. We instead ask whether evidence-inconsistent geometry can be repaired while identity, lifecycle, trajectory, size, time-to-collision, and hazard events remain fixed.

We propose HARP-3D, a physical compilation layer between a reconstructed scene and downstream task queries. It represents a world as

$$\mathcal{W}_t = (\mathcal{G}_t^{\mathrm{app}}, \mathcal{S}_t^{\mathrm{phys}}, \mathcal{A}_t, \mathcal{P}_t, \mathbf{q}_t^{\mathrm{val}}), \tag{1}$$

where appearance Gaussians and physical collision surfaces have separate semantics. The compiler uses multi-frame Actor alignment, ray free/occupied evidence, temporal support, and source provenance to select one of four local actions. Unsupported regions become UNKNOWN, never free by default, while Actors remain in the scene.

Our second component factorizes validity from hazard with paired counterfactuals. Changing a clean Actor from ordinary driving to a legal near-miss should not change its artifact score; injecting a duplicate shell or temporal pop should not change its hazard score. Our third component links physical state quality to downstream reliability. An Actor residual distribution is projected onto a candidate trajectory boundary, yielding a continuous cost whose conditional density supports arbitrary budgets and multiple horizons.

Our contributions are:

- a hazard-preserving physical compiler with explicit KEEP/PROJECT/COMPLETE/UNKNOWN semantics and separate appearance/collision state;
- a paired validity–hazard factorization protocol that measures cross-task leakage rather than relying on global

decorrelation;

- a continuous task-cost density and monotone multi-horizon reliability interface, evaluated under a frozen nuScenes-to-Argoverse 2 zero-shot protocol.

2. Related Work

Driving-world reconstruction. Multi-sensor driving datasets such as nuScenes [1] and Argoverse 2 [7] provide calibrated imagery, LiDAR, ego pose, maps, and tracked Actors. Neural scene graphs model dynamic objects explicitly [4], while 3D Gaussian Splatting provides a high-quality appearance representation [2]. Our work does not reinterpret Gaussian opacity as collision occupancy. It adds a separate, queryable physical surface with sensor provenance.

Uncertainty and reliability. Deep ensembles expose useful empirical predictive disagreement [3], and conformalized quantile methods can turn residuals into calibrated marginal intervals under appropriate assumptions [5]. Driving logs violate many simple exchangeability assumptions through scene correlation and domain shift. We therefore report empirical calibration explicitly, preserve claim boundaries, and use monotone structure only for runtime consistency—not as a formal road-safety guarantee.

Our distinction. Prior confidence, occupancy, and reconstruction-quality signals often mix physical invalidity, visibility, and behavioral difficulty. HARP-3D instead defines separate evidence inputs, paired interventions, and failure denominators so that an improvement cannot be obtained by deleting hazardous Actors or relabeling unknown space as free.

3. Problem Formulation

Let Actor i have an immutable semantic state

$$A_i = (ID_i, X_{i,0:H}, D_i), \tag{2}$$

containing identity, trajectory, and dimensions. Appearance primitives may be associated with A_i, but collision queries consume only an Actor-local physical surface S_i^{phys} and its known/unknown mask.

We separate two random variables. Artifactness a_i depends on sensor and geometric evidence E_i, physical surface state, and provenance:

$$a_i = P(\text{artifact} \mid E_i, S_i^{\text{phys}}, P_i). \tag{3}$$

Hazardness h_i depends on Actor dynamics, map context, and a candidate Ego trajectory τ:

$$h_i = P(\text{hazard} \mid X_{i,0:H}, \tau, \mathcal{M}). \tag{4}$$

The goal is paired conditional invariance, not unconditional statistical independence. For the same clean Actor, a legal safe-to-hazard intervention should preserve a_i. For the same behavior, a clean-to-artifact intervention should preserve h_i.

For task reliability, let $n_{\tau,t}$ be the local boundary normal, $d_{i,t}(\tau)$ the signed Actor clearance, and $\epsilon_{i,t}$ the Actor residual. We define

$$C(\tau, H) = \max_{i,t \le H} \frac{|n_{\tau,t}^\top \epsilon_{i,t}|}{\max(|d_{i,t}(\tau)|, \varepsilon)}. \tag{5}$$

The runtime interface returns $R(\tau, H, b) = P(C \le b)$ and enforces $\partial R/\partial b \ge 0$ and $\partial R/\partial H \le 0$.

4. Method

4.1. Actor-Preserving Physical Compilation

For each Actor, we transform LiDAR endpoints, rays, and associated primitives into a canonical Actor frame,

$$x_{i,t}^{\text{actor}} = T_{\text{actor},t}^{-1} T_{\text{ego},t} T_{\text{sensor}} x_t^{\text{sensor}}. \tag{6}$$

The fused representation stores surface support, observed-free intervals, temporal counts, local normals, viewing-direction diversity, and provenance. Compilation is deterministic and local:

- KEEP: stable sensor-supported surface;
- PROJECT: a near-surface primitive contradicts observed free space and is projected to supported geometry;
- COMPLETE: a local hole has sufficient multi-frame and multi-view support;
- UNKNOWN: evidence is missing, contradictory, or insufficient.

Unknown is a first-class state and never silently becomes free. All four actions preserve $(ID_i, X_{i,0:H}, D_i)$. Appearance primitives may attach to the repaired surface using center distance, covariance-normal alignment, depth, and visibility, but physical queries do not read Gaussian opacity.

4.2. Validity–Hazard Factorization

The validity encoder receives ray contradiction, surface residual, temporal support, provenance, visibility, and local geometry. The hazard encoder receives relative pose and velocity, TTC, route relation, map context, and interaction state. Two supervised heads are augmented with paired consistency losses:

$$\mathcal{L} = \mathcal{L}_{\text{artifact}} + \mathcal{L}_{\text{hazard}} + \lambda_{\text{pair}} \mathcal{L}_{\text{pair}} + \lambda_{\text{leak}} \mathcal{L}_{\text{leak}}. \tag{7}$$

Pairs span a four-quadrant atlas: valid-safe, valid-hazard, artifact-safe, and artifact-hazard. We measure hazard-from-validity and artifact-from-hazard probes in addition to task AUROC/AUPRC.

Multi-frame Actor alignment ray FREE/OCC evidence → Physical compiler KEEP / PROJECT COMPLETE / UNKNOWN → Validity–hazard paired factorization → Actor residuals analytic task query → Cost density monotone reliability

Figure 1. HARP-3D compiles sensor evidence into an Actor-preserving physical world, separates validity from hazard, and exposes task-conditioned reliability. Appearance Gaussians never serve directly as collision occupancy.

4.3. Actor Residuals and Boundary-Cost Density

For Actor i and horizon t, a low-capacity head predicts a 2D residual distribution $(\mu_{i,t}, \Sigma_{i,t})$. Analytic projection onto $n_{\tau,t}$ gives mean $n^\top \mu$ and variance $n^\top \Sigma n$. Equation 5 converts these residuals and deterministic clearance into a continuous trajectory cost.

We model $y = \log(1+C)$ with a small conditional Gaussian mixture,

$$p_\theta(y \mid z^{\text{val}}, z^{\text{haz}}, \tau, H) = \sum_k \pi_k \mathcal{N}(y; \mu_k, \sigma_k^2), \qquad (8)$$

which exposes a closed-form marginal CDF for any budget. A low-rank, input-conditioned dependence model combines the frozen marginals across horizons. This factorization separates learned Actor uncertainty from analytic task geometry.

4.4. Monotone Runtime Surface and SceneIR

The final runtime object is a frozen grid over horizon and cost budget with monotone interpolation. Budget CDF values are non-decreasing; prefix-horizon reliability is non-increasing. Groupwise isotonic maps may calibrate predefined conditions on a disjoint source fold, but do not change these structural constraints.

SceneIR stores Actor identities and transforms, physical surfaces, known/unknown masks, provenance, and validity fields. The AV2 adapter performs only coordinate, unit, timestamp, category, and sensor-opportunity normalization. It does not fine-tune, recalibrate, or change thresholds using AV2 quality.

5. Experiments

Data and protocol. We train and select models only on nuScenes. Argoverse 2 Sensor validation logs form a zero-shot external test. Before reading any method output, we sort the 150 validation UUIDs and select every fifth log, yielding 20 quantitative and 10 qualitative logs. No AV2 fine-tuning, post-hoc calibration, threshold selection, or failed-scene deletion is allowed. Final nuScenes and AV2 cohorts are evaluated exactly once after the method freezes.

Baselines. For physical compilation we compare naive reconstruction, a legacy validity filter, V6.7 inward-ray repair, and HARP-3D. For factorization we compare a shared encoder, a two-head shared trunk, independent encoders, and paired factorization. Reliability baselines include a scalar confidence, independent horizon classifiers, a continuous marginal density, and the full joint multi-horizon surface.

Metrics. Physical metrics include free-space violation, ray termination, point-to-surface error, LiDAR depth consistency, temporal jitter, ghost components, and surface completeness. Hazard preservation reports Actor/ID/lifecycle retention, trajectory and TTC shift, and event-count change. Factorization reports artifact/hazard AUROC and AUPRC, clean-hazard false-artifact rate, paired prediction shift, and cross-probe leakage. Reliability reports log likelihood, integrated Brier score, calibration error, Spearman correlation, monotonic violations, and runtime.

Zero-shot Actor-local surface evidence. Following the dynamic-Actor canonicalization boundary used by NeuRAD [6], we transform ego-frame AV2 LiDAR points inside each annotated cuboid into its Actor frame. Unlike appearance-oriented seed initialization, we never mirror points: every collision surfel must retain sensor and temporal support. We fuse two of every three frames at 0.12 m and reserve the remaining frames as unseen target geometry. Across 1,046 Actors and 1.075M stable surfels, fusion reduces the mean target-to-surface distance from 0.864 m to 0.131 m and raises recall at 0.20 m from 26.16% to 85.72% (Table 1).

The paired corruption atlas contains clean, observed-free ghost, duplicate-shell, temporal-flicker, and supported-hole probes without changing Actor identity, trajectory, size, or hazard. It yields 100.0% artifact recall, 0.00% clean-hazard false-artifact rate, and 100.0% Actor/hazard retention. These perfect paired scores verify the deterministic compiler contract and input separation; they are not reported as natural-artifact generalization.

Four-action physical compilation. We next freeze one single-frame query per Actor and reserve all later held-out frames as target-only geometry. The compiler reads only the build surface and query evidence. A nearest-canonical projection initially reduces Chamfer to 0.168 m,

Figure 2. The first metadata-frozen AV2 main case. From left: synchronized RGB with observed Actor returns, paired synthetic artifacts, the four-action ray-certified output, and motion-compensated sparse camera depth. Camera/crop selection does not read RGB appearance or method quality. This is a calibrated evidence overlay, not photorealistic reconstruction.

AV2 Actor surface	Dist. (m) ↓	Recall (%) ↑
Single observed frame	0.864	26.16
Multi-frame canonical surfels	**0.131**	**85.72**

Table 1. Zero-shot held-out LiDAR support on the frozen 20-log AV2 quantitative cohort (1,046 Actors, including 241 hazardous Actors). The comparison uses real sensor points; paired synthetic corruptions are evaluated separately as a compiler contract.

Surface	Free (%) ↓	Recall (%) ↑	CD (m) ↓
Single-frame input	100.0	53.36	0.254
Nearest canonical	84.91	**62.60**	**0.168**
Ray-certified	**0.00**	59.56	0.175

Table 2. Frozen zero-shot AV2 quantitative result. Nearest-surface projection fits geometry best but violates matched-ray free space; ray-certified projection accepts a small Chamfer cost. Later held-out frames are target-only.

but ray-specific evaluation reveals 84.91% early terminations in the paired observed-free interval. Our final operator is ray-certified: a primitive with direct LiDAR provenance projects to its matched observed termination; without one it becomes UNKNOWN. Across 433 quantitative Actors (147 hazardous), this reduces paired free-space violations from 100.0% to 0.00%, target depth error from 0.207 m to 0.154 m, and Chamfer from 0.254 m to 0.175 m, while raising ray-termination consistency from 56.94% to 64.38% (Table 2). Actor identity, trajectory, extent, and hazard labels are retained for 100.0% of Actors. The zero free-space rate is a paired contract result; depth, ray termination, and Chamfer use independent target-only LiDAR.

Frozen camera evidence and an exposed boundary. We project each pre-registered qualitative Actor back into synchronized AV2 ring-camera frames using the official calibration and motion compensation. Camera identity maximizes only the number of visible query LiDAR returns, with a fixed camera-order tie break; RGB is decoded afterward and every case is retained. All 30 cases have calibrated RGB, sparse depth, four-action, and before/after video evidence; the median/minimum visible query counts are 82/14 (Fig. 2). Importantly, the frozen set exposes a limit: although mean Chamfer improves, 17.19% of all 634 Actors worsen individually. Thus P3 establishes aggregate zero-shot geometry, not a per-Actor repair certificate; this motivates a selective "repair or abstain" validity head trained and calibrated only on nuScenes.

Qualitative protocol. Main-paper and supplement scenes are frozen from metadata and registered failure strata before rendering method outputs. Panels show RGB, LiDAR, physical surface, ray evidence, compiler action, and the repaired world. We report all frozen cases, including failures, rather than selecting only visually favorable examples.

6. Limitations

HARP-3D is not a formal guarantee of real-road safety. Sparse or one-sided sensing can leave large regions unknown, and deterministic completion is intentionally conservative. Actor cuboids and tracked identities may themselves be wrong. Empirical isotonic calibration under scene correlation does not provide distribution-free conditional coverage, and nuScenes-to-AV2 transfer may fail through sensor opportunity, interface, representation, dynamics, or calibration shift. The current physical compiler targets Actor-local surfaces; static-scene topology and deformable Actors remain incomplete. Finally, preservation of identity, trajectory, and size guarantees only hazard functions that depend on those variables; appearance-dependent human judgments may still change.

Method	Free-space ↓	Surface error ↓	Actor retention ↑	Clean-hazard FA ↓	Brier ↓	AV2 zero-shot ↑
Naive reconstruction	TBD	TBD	TBD	TBD	TBD	TBD
V6.7 inward-ray	TBD	TBD	TBD	TBD	TBD	TBD
HARP-3D	TBD	TBD	TBD	TBD	TBD	TBD

Table 3. Paper-facing result contract. Values remain TBD until the corresponding frozen milestone completes. Metrics will be split into full physical, factorization, and reliability tables in the results draft.

7. Conclusion

We formulate driving-world compilation around a simple boundary: validity is not difficulty. HARP-3D repairs evidence-inconsistent physical geometry while preserving legitimate hazardous Actors, factorizes artifactness from hazardness with paired interventions, and exposes downstream reliability through a continuous multi-horizon cost distribution. The resulting SceneIR keeps appearance, collision state, unknown space, provenance, and task reliability explicit. The remaining work is empirical: completing the canonical-surface milestones, executing the frozen zero-shot protocol, and replacing all draft placeholders without expanding the claim boundary.

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